Advanced Theory — Module 06: Learning — Study Questions

1. Define model evidence (marginal likelihood). Why does it automatically balance fit against complexity?
2. What is the Occam window? Why are models that are too simple or too complex penalized?
3. How are Bayes factors computed? What thresholds indicate strong evidence?
4. How does free energy approximate model evidence for model selection?
5. What is Bayesian Model Reduction? How does it avoid refitting reduced models?
6. Derive the BMR formula under the Laplace approximation. What terms appear?
7. How does BMR formalize sleep-dependent parameter pruning?
8. What is structure learning? How does it differ from parameter learning?
9. Compare score-based, constraint-based, and BMR-based structure learning. Pros and cons?
10. How do Dirichlet process priors enable learning the number of hidden states?
11. What is a hierarchical Bayesian model? How do hyperparameters set priors?
12. What is Empirical Bayes? How does it learn priors from data?
13. What is Parametric Empirical Bayes? How does it implement group-level inference?
14. What is model space topology? How does it define neighborhoods between models?
15. How does greedy BMR search work? What is its stopping criterion?
16. What is reversible jump MCMC? Why is it needed for structure learning?
17. How does BMS relate to hypothesis testing? What are the advantages over classical frequentist testing?
18. How does group-level BMR enable neuroscientific conclusions about "typical" brain connectivity?
19. What is the relationship between BMS and the minimum description length (MDL) principle?
20. Critically evaluate: Does BMS + BMR provide a complete framework for scientific discovery, or are there situations where it fails?